Q1: compute a^b for integers > 0.

Basic way:

```
int r = 1;
for (int i = 0; i < b; i++)
    r *= a;
```

$$\underbrace{a * a * \cdots * a}_{b \text{ times}}$$

and we want $(a^b \% n)$

What if a, b are <u>huge</u>? (1000 digits...)

Hint: can you think of any very large value of e for which $a^e \% n$ is fast/easy to compute?

$$a \to a^2 \to a^4 \to a^8 \to a^{16} \to \cdots a^{2^\ell}$$

in $\approx \ell$ multiplications

(we always do $\% n$ even if I don't say it)

What about an arbitrary exponent $b \neq 2^\ell$?

Easy: write $b = \sum_{i=0}^{\ell} x_i 2^i$ $\quad x_i \in \{0,1\}$ (binary rep of b)

So $a^b = a^{\left(\sum_{i=0}^{\ell} x_i 2^i\right)} = \prod_{x_i=1}^{\ell} a^{2^i}$

Say $b = 19 = 11001$

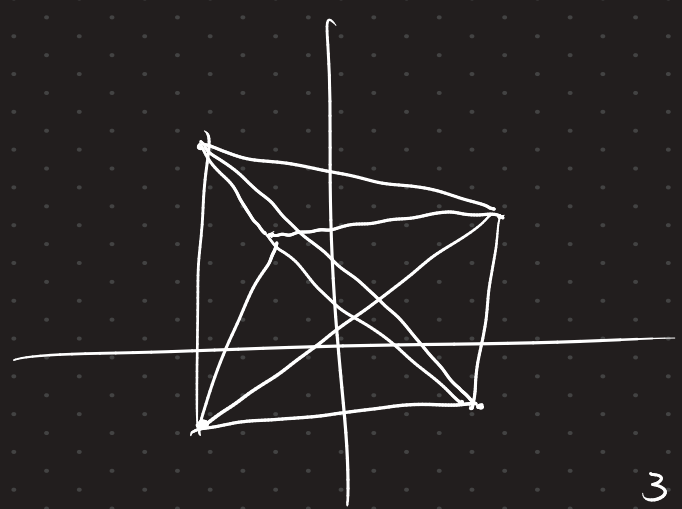
$$a^1 \to a^2 \to a^4 \to a^8 \to a^{16} \to \cdots a^{2^\ell}$$

$$a^1 \cdot a^2 \cdot a^{16} = a^b$$

all are efficient to compute!

look up "square and multiply" exponentiation.

Q3

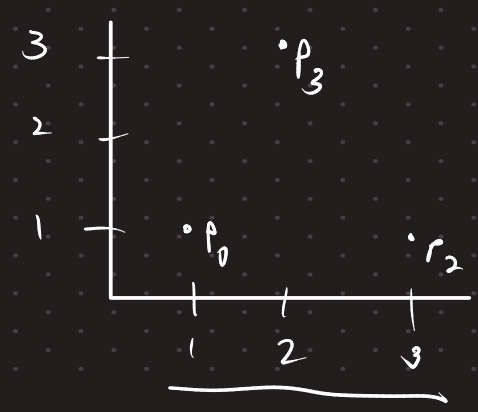


Representation: 2 vectors.

$p_0 \quad p_1 \quad p_2$

$X = \{1, 3, 2\}$

$Y = \{1, 1, 3\}$



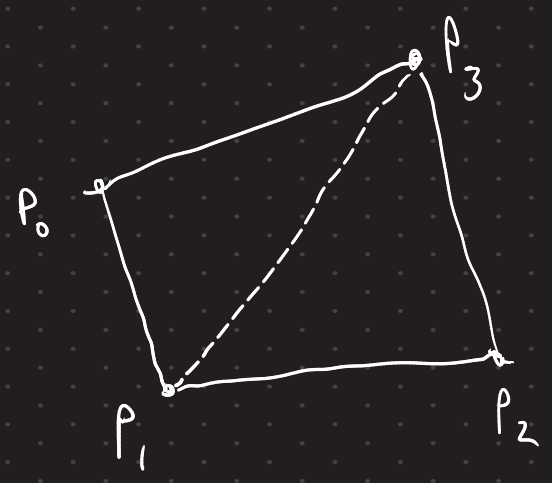
Recall the 2-set sum problem:

V a vector of int, t a "target".

Does there exist indexes $i \neq j$ s.t.

$$\underline{V[i] + V[j] == t}\ ?$$

Brute force solution: go through all (i,j) and check

Q3: same, but keep track max distance.

What about checking distance of $\underline{(P_{i-1}, P_i)}$?